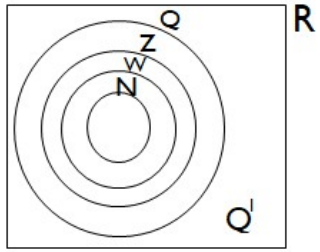
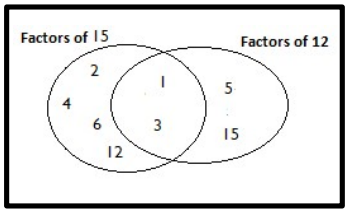
BASIC 9

STRAND 1: NUMBER

Sub-strand 1: Number and Numeration System

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.1.1 Apply the understanding of place value in solving real life problems involving integers of any size, rounding this to given decimal places and significant figures	**B9.1.1.1.1 Express integers to a given number of significant and decimal places**	**Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)**
	E.g.1. Express integers to a number of significant figures. (i) 857,386,321 -five significant figures -four significant figures -three significant figures.	
	E.g.2. Express decimal numbers to a given number of decimal places. (i) Write 98745.9674 correct to -three decimal places -two decimal places -one decimal place	
	B9.1.1.1.2. Use knowledge and understanding of place value to solve real life problems	
	E.g.1. Create and solve a real-life problem or a story problem and write the answer in standard form. (I) I am a 6-digit number. My first digit is 5 more than the last digit, but 2 less than my second digit. My second digit is the third multiple of 3, while my fourth digit is the second multiple of 3. My third digit is the quotient when the fourth digit is divided by my last digit. However, my fourth and fifth digits are consecutive numbers. What number am I?	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
	Think second digit: 3x3=9 fourth digit: 2x3=6 first digit: 9-2=7 last digit: 7-5=2 fifth digit: 6-1=5 third digit: 6□□□□ So, the number is 793652 = 7.93652×10^5	
	E.g.2 Create similar real story problems and solve	
B9.1.1.2 Demonstrate an understanding of the relationship between members of the rational number system and solve real life problems involving union and intersection of three sets	**B9.1.1.2.1 Solve problems on relationship between members of the rational number system using knowledge and understanding of the concept of union and intersection of two sets.**	**Communication and Collaboration (CC)** **Creativity and Innovation (CI)**
	E.g. 1 Use sets diagrams to show the relationship among the Real numbers namely	
	-Irrational numbers (Q^I) -Rational numbers (Q) - Integers (Z) -Whole numbers (W) -Natural or Counting numbers (N)	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.1.2 Demonstrate an understanding of the relationship between members of the rational number system and solve real life problems involving union and intersection of three sets	E.g. 2 Write the factors of 12 and 15 and represent them on a Venn diagram. 12={1, 2, 3, 4, 6, 12} 15={1, 3, 5, 15} 	**Communication and Collaboration (CC)**
	B9.1.1.2.2 Apply the concept of sets to solve problems on relationship between members of rational number system and solve real life problems involving union and intersection of two sets	
	E.g.1 Create and solve real life problems to show the union and intersection of two sets. i. There are 80 farmers in a certain village who grow either maize or beans. Fifty of them grow beans while sixty grow maize. If each farmer grows at least one of the two crops, represent the information on a Venn diagram and hence find the number of farmers who grow: a. both crops. b. only one crop.	

STRAND 1: NUMBER

Sub-strand 2: Number Operations

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B.9.1.2.1 Apply mental mathematics and properties to determine answers for addition and subtraction of basic facts.	**B9.1.2.1.1 Multiply and divide given numbers by powers of 10 including decimals and benchmark fractions** E.g.1. Recall multiplication facts up to 144 and related division facts. E.g.2. Recall decimal names of given benchmark fractions converted to decimals or percentages (and vice versa) E.g. 3. Find the product of a given decimal number when it is multiplied by 10, 100, 1000, $\frac{1}{10}$, $\frac{1}{100}$, $\frac{1}{1000}$, etc.	**Creativity and Innovation (CI)**
	B.9.1.2.1.2 Demonstrate the ability to determine commutative properties of addition and multiplication. E.g1. Recognise that for any two numbers **a** and **b**; i. a + b = b + a i.e. 25 + 32 = 32 + 25 = 57 ii. a × b = b × a i.e. 17 × 8 = 8 × 17 = 136 **B9.1.2.1.3 Use the associative property of addition and multiplication.**	
	E.g1. Recognise that for any three numbers **a**, **b** and **c**; i. a + (b + c) = (a + b) + c or a + (b + c) = (a + c) + b i.e. 15 + (6 + 9) = (15 + 6) + 9 = 30 ii. (a × b) × c = a × (b × c) i.e. (12 × 5) × 4 = 12 × (5 × 4) = 240	

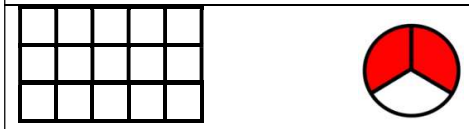
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
	B9.1.2.1.4 Use the distributive property in solving problems.	
	E.g1. Recognise that for any three numbers **a**, **b** and **c**; i. a × (b + c) = (a × b) + (a × c) i.e. 5 × (10 + 7) = (5 × 10) + (5 × 7) = 85 ii. a × (b - c) = (a × b) - (a × c) i.e. 5 × (10 - 7) = (5 × 10) - (5 × 7) = 15	
B9.1.2.2 Apply the understanding of addition, subtraction, multiplication and division of decimal numbers to solve problems, and round answers to given decimal places and significant figures	**B9.1.2.2.1 Solve operations involving addition, subtraction, multiplication and division using word problems.**	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)**
	E.g. Create and solve story problems involving a combination of two or more of the basic operations. (×, ÷, ─,+). i) A trader sells oranges from two baskets, A and B. Basket A contained 85 oranges and she sold 48. She sold 59 oranges from basket B and was left with the same number of oranges as in Basket A. How many oranges were originally in Basket B?	
	B9.1.2.2.2 Solve word problems involving the four basic operations and round the answers to the nearest two decimal figures or to some significant figures.	
	ii) The price of a jacket is three times that of a shirt. The price of a jacket is GH₵560.65. Mr Mensa bought two of the jackets and four shirts for his twin sons. Calculate the total amount Mr Mensa paid for the items, correct your answer to: α) two decimal places β)three significant figures	

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr><td rowspan="2">B9.1.2.4 Demonstrate understanding of surds as real numbers, the process of adding and subtracting of surds as well as determining (using a calculator) the approximate square root of a number that is not a perfect square.</td><td>B9.1.2.4.1 Identify simple and compound surds.
E.g.
i. $\sqrt{2}$, ii. $7\sqrt{3}$, iii. $2\sqrt{5}$
iv. $(\sqrt{3}+\sqrt{7}-\sqrt{5})$
B9.1.2.4.2 Explain the identities/rules of surds
Rule 1
$\sqrt{a\times b}=\sqrt{a}\times\sqrt{b}$
Rule 2
$\sqrt{\frac{a}{b}}=\frac{\sqrt{a}}{\sqrt{b}}$
Rule 3
$\frac{b}{\sqrt{a}}=\frac{b}{\sqrt{a}}\times\frac{\sqrt{a}}{\sqrt{a}}=\frac{b\sqrt{a}}{a}$
Rule 4
$a\sqrt{c}\pm b\sqrt{c}=(a\pm b)\sqrt{c}$
Rule 5
$\frac{c}{a+b\sqrt{n}}=\frac{c}{a+b\sqrt{n}}\times\frac{a-b\sqrt{n}}{a-b\sqrt{n}}$
Rule 6
$\frac{c}{a-b\sqrt{n}}=\frac{c}{a-b\sqrt{n}}\times\frac{a+b\sqrt{n}}{a+b\sqrt{n}}$</td><td rowspan="2">Critical Thinking and Problem solving (CP)
Communication and Collaboration (CC)
Creativity and Innovation (CI)</td></tr>
<tr><td>B9.1.2.4.3 Simplify given surds
E.g.1 Simplify:
i. $\sqrt{27}$
i. $\sqrt{72}$
ii. $\frac{\sqrt{8}}{16}$
iii. $\frac{\sqrt{12}}{121}$
iv. $(\sqrt{2})^2$</td><td>)</td></tr>
</table>

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.2.4 Demonstrate understanding of surds as real numbers, the process of adding and subtracting of surds as well as determining (using a calculator) the approximate square root of a number that is not a perfect square.	**B9.1.2.3.4 Approximate the square roots of non-perfect squares with calculators/tables.** E.g.1 Square roots of non-perfect squares i. $\sqrt{2}$ ii. $\sqrt{5}$ iii. $\sqrt{12}$ iv. $\sqrt{30}$	**Personal Development and Leadership (PL)** **Cultural Identity and Global Citizenship (CG)**

STRAND 1: NUMBER

Sub-strand 3: Fractions, Decimals and Percentages

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.3.1 Apply the understanding of operations on fractions to solve problems involving fractions of given quantities and round the results to given decimal and significant places	**B9.1.3.1.1 Review fractions and solve problems involving basic operations on fractions**	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Personal Development and Leadership (PL)** **Creativity and Innovation (CI)**
	E.g. 1. Review the concept of fraction.  i. Shade the fraction of squares in the rectangle that is equal to the shaded portion of the circle. ii. Write down 3 fractions equivalent to $\frac{2}{5}$ iii. Express the fraction $\frac{15}{10}$ in its simplest form: $\frac{15}{10}$ iv. Express $\frac{12}{5}$ as a mixed number: $\frac{12}{5}$ v. Express $2\frac{5}{9}$ as an improper fraction: $2\frac{5}{9}$	
	E.g. 2. Review the basic operations on fractions. i. Adding and subtracting fractions: Work out answers to the following: a) $\frac{3}{4}+\frac{7}{8}$ b) $1\frac{1}{2}+\frac{4}{5}-\frac{5}{6}$ ii. Multiplying and dividing fractions. Work out answers to the following: a) $\frac{2}{3}\times\frac{3}{4}-\frac{3}{8}$ b) $\frac{5}{8}\div 2\frac{1}{2}+\frac{2}{3}$	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.3.1 Apply the understanding of operations on fractions to solve problems involving fractions of given quantities and round the results to given decimal and significant places	**B9.1.3.1.2 Add and/or subtract, multiply and/or divide given fractions, using the principle of order of operations including the use of the BODMAS or PEMDAS rule, and apply the understanding of these to solve problems.** E.g. 1. Use the order of operations (BODMAS or PEDMAS) to simplify whole number expressions with more than two operations. PEDMAS is Parenthesis, Exponents, Multiply/Divide (going from left to right)and Add/subtract (going from left to right). i. $3^4 \div 3^2 + 40 - 2^3 \times 3^2 \div 9$ ii. $18 \div 6 \times (4 - 3) + 6$ iii. $18 \div 3^2 \times (4 - 3) \times 10$ E.g. 2. Use the order of operations (BODMAS or PEDMAS) to simplify fractions with more than two operations. a) $\frac{2}{3} \times \frac{3}{4} - \frac{5}{8} \div 2\frac{1}{2}$ b) $\frac{3}{4} \div \frac{3}{8} + (\frac{4}{5} - \frac{1}{2})$ c) $(\frac{3}{4} + \frac{5}{8}) \times \frac{4}{11} - \frac{1}{2}$	**Creativity and Innovation (CI)** **Personal Development and Leadership (PL)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.3.1 Apply the understanding of operations on fractions to solve problems involving fractions of given quantities and round the results to given decimal and significant places	**B9.1.3.1.3. Review word problems involving basic operations on fractions.** E.g. 1. Solve word problems based on fractions. i. A test is made up of 20 questions, how many questions must you answer correctly to get a score of 80%? ii. By what percentage was a television set reduced if it was marked GH₵2,250 and sold for GH₵2,025? iii. In an election involving two contestants, one candidate claimed 52% of the votes, while the other candidate claimed 2,681 votes. If 5000 people voted, how do you know the election results are invalid? iv. Esi and Fusena prepared orange drink by mixing orange squash and water. Esi's drink was made of $\frac{3}{8}$ orange squash and Fusena's was made up of $\frac{2}{5}$ orange squash. Whose drink tasted stronger of orange?	**Creativity and Innovation (CI)** **Cultural Identity and Global Citizenship (CG)**

STRAND 1: NUMBER

Sub-strand 4: Number: Ratios and Proportion

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.4.1 Apply the understanding of ratio, rate and proportions to solve problems that involve rates, ratios, and proportional reasoning and use it to solve real-world mathematical problems.	**B9.1.4.1.1 Represent proportional relationships by equations.**	**Critical Thinking and Problem solving (CP)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**
	E.g.1 If total cost **(t)** is proportional to the number of items **(n)** purchased at a constant price **(p),** the relationship between the total cost and the number of items can be expressed as **t = pn**.	
	B9.1.4.1.2 Use proportional relationships to solve multistep ratio and percent problems, examples: simple interest, tax, discount and commissions, NHIL, depreciation, insurance, etc.	
	E.g.1 Solve problems on simple interest. i. A girl deposited GH₵ 4500 at the bank at a rate of 3% per annum for three years. Find the simple interest. What is the amount at the end of the fifth year?	
	E.g.2 Solve problems on tax (VAT). i. The VAT rate of Ghana is 12.5%. A man bought an item at GH₵ 4500.00, VAT inclusive. Calculate: b) the basic cost of the item. c) the VAT paid by the man.	
	E.g.3 Solve problems on discount. **i.** If a car costs GH₵ 80,500.00,what is its new value if there is a discount of 10%? **E.g.4 Solve problems on commission.** ii. A car agent's commission on the sale of a car is $3\frac{1}{2}$%. Calculate the commission on a car sold for GH₵68,000.00.	

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr><td>B9.1.4.1 Apply the understanding of ratio, rate and proportions to solve problems that involve rates, ratios, and proportional reasoning and use it to solve real-world mathematical problems.</td><td>E.g.5 solve problems involving depreciation.
The value of a mobile phone depreciates at the following rate:

<table><tr><th>Year of manufacture</th><th>Depreciation on the original value</th></tr><tr><td>In the first year</td><td>5%</td></tr><tr><td>In the second year</td><td>10%</td></tr><tr><td>In the third year</td><td>15%</td></tr><tr><td>In the fourth year</td><td>22%</td></tr></table>
The original value of the mobile phone is GH₵ 1800.00. Find the value of the mobile phone at the end of each of the first four years.</td><td>Critical Thinking and Problem solving (CP)

Creativity and Innovation (CI)

Personal Development and Leadership (PL)

Digital Literacy (DL)</td></tr>
<tr><td></td><td>E.g.6 Solve problems involving NHIL.

i. The NHIL inclusive price of a television set is GH₵1200.00. If the NHIL is charged at a rate of 2.5%, find
b) the cost of the television set (NHIL exclusive).
c) the NHIL charged.</td><td></td></tr>
<tr><td></td><td>E.g.7 Solve problems involving insurance.

i. Kofi Mereku insured his house and paid a premium of GH₵ 30,000.00. If the insurance company fixed the rate at 5% of the value of the house, calculate the insured value of the house.</td><td></td></tr>
<tr><td></td><td>B9.1.4.1.3 Use knowledge of rates and proportional reasoning to solve problems involving SSNIT benefits and contributions.</td><td></td></tr>
<tr><td></td><td>E.g.1 Describe the obligations of the employer/employee and the contribution rates.
<table><tr><th></th><th>Act 766</th><th>PNDC Law 247</th></tr><tr><td>Employer</td><td>13.0% of basic salary</td><td>12.5 of basic salary</td></tr><tr><td>Worker</td><td>5.5% of basic salary</td><td>5.0% of basic salary</td></tr><tr><td>Total</td><td>18.5% of basic salary</td><td>17.5% of basic salary</td></tr></table></td><td></td></tr>
</table>

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr><td>B9.1.4.1 Apply the understanding of ratio, rate and proportions to solve problems that involve rates, ratios, and proportional reasoning and use it to solve real-world mathematical problems.</td><td>E.g.2 Calculate employee/employer contributions to SSNIT under Act 766.

A worker’s basic monthly salary is GH₵3,256.50.

b. Calculate the SSNIT contributions under Act 766;

i) by the employer
ii) by the employee

What is the total SSNIT contributions at the end of every month?</td><td>Critical Thinking and Problem solving (CP)</td></tr>
<tr><td></td><td>E.g.3 Calculate employee/employer contributions to SSNIT under PNDCL 247.

Mr Bediako’s monthly SSNIT contribution under PNDCL 247 is GH₵440.54. How much does his employer contribute to SSNIT on his salary? Hence, calculate his basic salary per month.</td><td></td></tr>
<tr><td></td><td>E.g.4 Calculate employee benefits from SSNIT under Act 766.

The table shows the pension rights for SSNIT contributors under Act 766

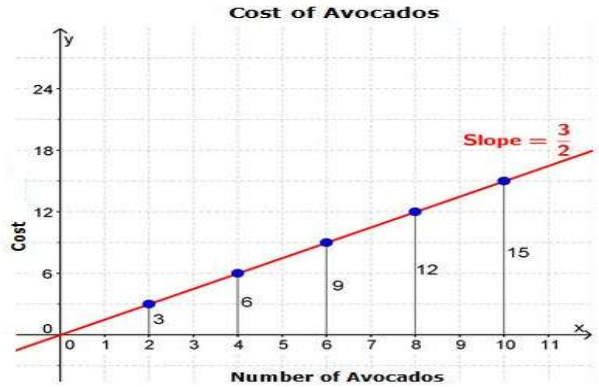
<table>
<tr><td>Years of contributions</td><td>15</td><td>16</td><td>17</td><td>18</td><td>…</td><td>30</td><td>31</td><td>32</td><td>…</td><td>35</td></tr>
<tr><td>Pension Rights (%)</td><td>37.500</td><td>38.625</td><td>39.750</td><td>40.875</td><td>…</td><td>54.375</td><td>55.500</td><td>56.625</td><td>…</td><td>60.000</td></tr>
</table>

a. Mr Addai retired at age 60 last year after working and contributing for 20years. If the average of his best salary for 3 years (36 months) over the 20-year period was GH₵15,000.00, calculate his full pension under the National Pension Act 2008, (Act 766).

Calculation for full pension
Qualifying age = 60years
Average best 3years’ salary = GH₵15,000
Pension right for 20years = 43.13% (refer to the table on Pension Rights above)

Annual pension to Mr. Addai = 43.13/100 × 15,000</td><td></td></tr>
</table>

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.1.4.1 Apply the understanding of ratio, rate and proportions to solve problems that involve rates, ratios, and proportional reasoning and use it to solve real-world mathematical problems.	= GH₵6,469.50 Monthly pension to Mr Addai = $\frac{\text{GH₵6,469.5}}{12}$ =**GH₵ 539.13** b. A worker contributed for $7\frac{1}{2}$years before being rendered incapacitated. If the best salary for over the 3-year (36 months) period was GH₵ 8,450.40, calculate the **invalidity benefit** for the worker. **c.** Mr Mensah's total SSNIT contribution stood at GH₵ 112,426.29 at the time of his demise. Calculate his **survivor's benefit** if the current interest rate is 15%.	**Critical Thinking and Problem solving (CP)**
	E.g.5 Calculate employee benefits from SSNIT under PNDCL 247. The table shows the pension rights for SSNIT contributors under PNDCL 247. **Years of contributions** | 20 | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 | … | 40 & above **Pension Rights (%)** | 50.0 | 51.5 | 53.0 | 54.5 | 56.0 | 57.5 | 59.0 | 60.5 | 62.0 | 63.5 | 65.0 | | 80.0 Mr Bema, a history teacher at Academicals Senior High School, retired in 2009 after 25 years of service. Throughout this 25-year period he had been an active contributor to the SSNIT Pension Scheme. As the student who has learnt about social security, you are to help Mr Bema to calculate his annual pension using his best three years' salary of GH₵19,500. **Calculation for full pension** Qualifying age = 60years Average best 3years' salary = GH₵19,500 Pension right for 25years = 57.5% (refer to the table on Pension Rights above) Annual pension to Mr. Bema = $\frac{57.5}{100} \times \text{GH₵19,500} = \frac{\text{GH₵11,212.5}}{12} = \mathbf{GH₵934.38}$	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
	B9.1.4.1.4 Recognise and graph proportional relationships, interpreting the unit rate as the slope of the graph and use these to solve problems.	**Digital Literacy (DL)** **Creativity and Innovation (CI)**
	E.g.1 The graph below shows the cost of avocados.  The unit rate, from the data, is ₵1.50 per avocado, which is the same as the slope of the line connecting the data points $(\frac{3}{2})$. i. From the graph, how much does eight avocados cost? ii. Also, using the graph how much does 15 avocados cost?	

STRAND 2: ALGEBRA

Sub-strand 1 Patterns and Relations

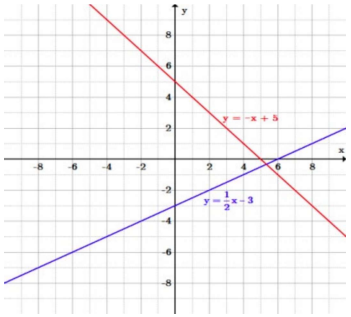
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<tr>
<td rowspan="2">B9.2.1.1 Demonstrate the ability to construct tables of values for pairs of linear relations, graph the relations in a number plane and determine the intersection of the lines to solve simultaneous linear equations.</td>
<td colspan="2">B9.2.1.1.1 Construct a table of values for two linear relations and graph the relation</td>
<td rowspan="2">Critical Thinking and Problem solving (CP)

Communication and Collaboration (CC)

Creativity and Innovation (CI)

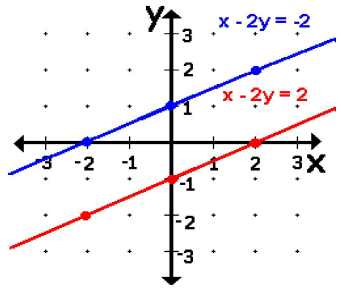
Cultural Identity and Global Citizenship (CG)</td>
</tr>
<tr>
<td>E.g.1Construct a table of values for two linear relations and draw the graphs of the relations

Copy and complete the table of values for the relations $y_1 = -x + 5$; and $y_2 = \frac{1}{2}x - 3$ for x from -4 to 3.
<table>
<tr><td>x</td><td>-3</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr>
<tr><td>$y_1 = -x + 5$</td><td>8</td><td></td><td></td><td></td><td>4</td><td></td><td></td></tr>
<tr><td>$y_2 = \frac{1}{2}x - 3$</td><td></td><td>-4</td><td></td><td></td><td></td><td></td><td>-1.5</td></tr>
</table>
</td>
<td>E.g.2 Draw graph for two linear relations.

</td>
</tr>
<tr>
<td></td>
<td>E.g.3Construct a table of values for two linear relations.

(i) Copy and complete the table of values for the relations $x - 2y = -2$ and $x - 2y = 2$ for x from -2 to 2.
<table>
<tr><td>x</td><td>x</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td></tr>
<tr><td>$x - 2y = -2$</td><td>y= $(x + 2)/2$</td><td>0</td><td></td><td></td><td></td><td>2</td></tr>
<tr><td>$x - 2y = 2$</td><td>$= (x - 2)/2$</td><td></td><td>-1$\frac{1}{2}$</td><td></td><td></td><td>0</td></tr>
</table>
</td>
<td>E.g.4 Draw a graph for two linear relations.

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</tr>
</table>

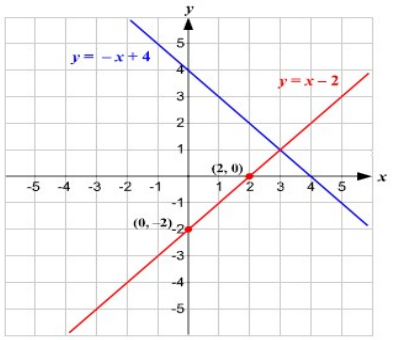
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<tr><td rowspan="2">B9.2.1.1 Demonstrate the ability to construct tables of values for pairs of linear relations, graph the relations in a number plane and determine the intersection of the lines to solve simultaneous linear equations.</td><td colspan="2">B9.2.1.1.2 Use graphs of two linear relations to determine subsequent missing elements in ordered pairs of the relation.</td><td></td></tr>
<tr><td>E.g.1 Find the missing elements of ordered pairs on graphs of two linear relations.

The graph below is drawn from two linear relations:

$y = -x + 4$

$y = x - 2$

i. Determine the coordinates for the intersection of the two lines.
ii. Determine the corresponding values for y for both straight lines if x = -1.
iii. Use the graph to find the values for y for the two relations.

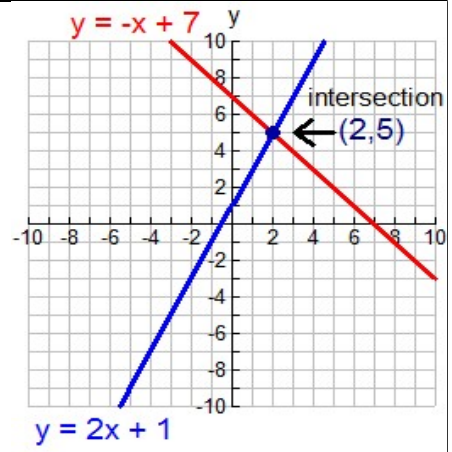
<table><tr><td>x</td><td>6-3</td><td>7-2</td><td>8-1</td><td>90</td><td>1</td><td>2</td></tr><tr><td>$y = -x + 4$</td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>$y = x - 2$</td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table></td><td> 
</td><td>Creativity and Innovation (CI)</td></tr>
</table>

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr><td rowspan="2">B9.2.1.1 Demonstrate the ability to construct tables of values for pairs of linear relations, graph the relations in a number plane and determine the intersection of the lines to solve simultaneous linear equations.</td><td colspan="2">B9.2.1.1.3 Use graphs to solve equations involving two linear relations.</td><td rowspan="2">Creativity and Innovation (CI)</td></tr>
<tr><td>E.g.1 Solve two linear equations simultaneously using the graph shown.

i. Solve the following equations simultaneously using a graph.
$y = -x + 7$

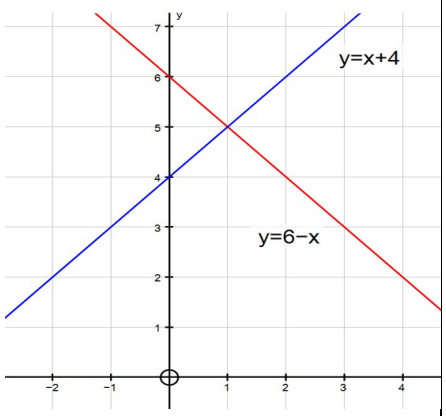
$y = 2x + 1$

Hint: Draw the graph and find the coordinates for the intersection of the two lines.

In the graph shown the values of (x, y) = (2, 5)</td><td> 
</td></tr>
<tr><td></td><td>E.g.2 Solve two linear equations simultaneously using the graph.

From the graph, determine the values of x and y that makes the linear equations true.

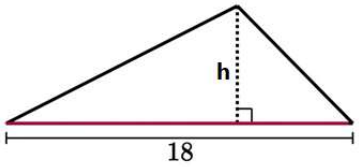
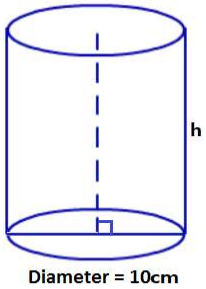
$y = x + 4$

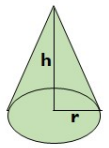
$y = 6 - x$</td><td> 
</td><td></td></tr>
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STRAND 2: ALGEBRA

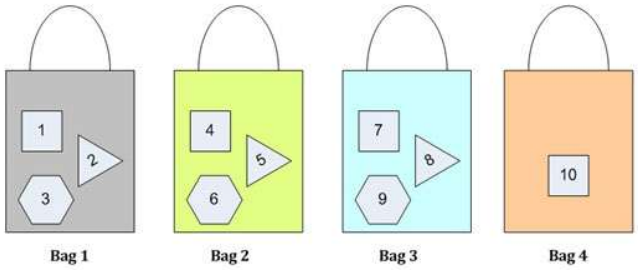
Sub-strand 2 Algebraic Expressions

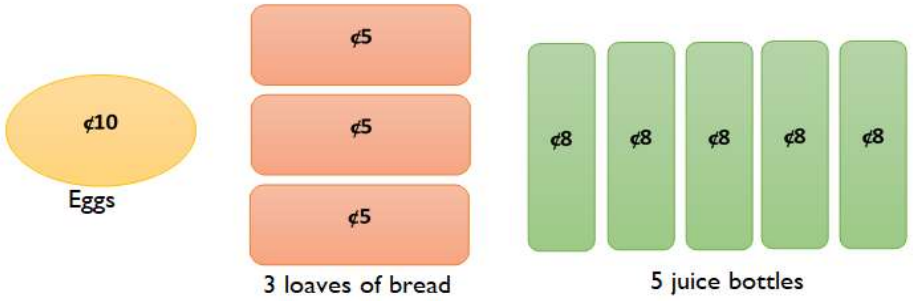
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**B9.2.2.1.1 Perform change of subject of a given formula and use it to solve problems.** **E.g.1** Perform change of subject for given formulae (i) Make x the subject of the following formulae 1) $q = x + 7$ 2) $r = x - 3$ 3) $5x = s$ 4) $\frac{3x+1}{2} = h$ 5) $3z = \frac{x}{4} + 1$	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Digital Literacy (DL)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	E.g.2 Use the concept of change of subject to solve problems involving formulae i. The area of a rectangle is 24cm^2. If the length is 8cm, find the value of the width. ii. The formula for calculating the area of a circle is given as πr^2. If a circle has an area of $154cm^2$, what is its radius?$[take\ \pi = \frac{22}{7}]$ iii. The triangle below has an area of 54cm^2. Find the value of the height of the triangle. iv. The cylinder below has a volume of 330cm^3. Find the value of the height of the cylinder. $[take\ \pi = \frac{22}{7}]$  	**Creativity and Innovation (CI)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**B9.2.2.1.2 Substitute values into given formulae to evaluate it and use it to solve problems.** **E.g.1** i. Find the value of $(\boldsymbol{x-b})^2-3(\boldsymbol{x-b})\ \boldsymbol{if x=2\ and\ b=-5}$ ii. Make $\boldsymbol{k}$ the subject of the formula: $\frac{\boldsymbol{1}}{\boldsymbol{n}}=\sqrt{(\frac{\boldsymbol{k^2+a^2}}{\boldsymbol{hg}})}$ If $\boldsymbol{n}=\frac{8}{5},\ a=3,\ h=2,\ g=32,$ find the value of k. iii. The formula for finding the volume of the shape below is given as $\frac{1}{3}\pi r^2 h$. Find the volume if r = 7, h = 21, a  	**Creativity and Innovation (CI)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**B9.2.2.1.3 Factorise expressions that have simple binomial.** **E.g.** Factorise the following expressions. i. $3x + 4xy = x(3 + 4y)$ ii. $12ab + 16b = 4b(3a + 4)$ iii. $-13xy + 39x = -13x(y-3)$ iv. $5y-2y^2+3y=-3y+3y$ v. $8y-2y^2= 2y(4-y)$ vi. $-6x+12=-3(2x-4)$	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Digital Literacy (DL)**

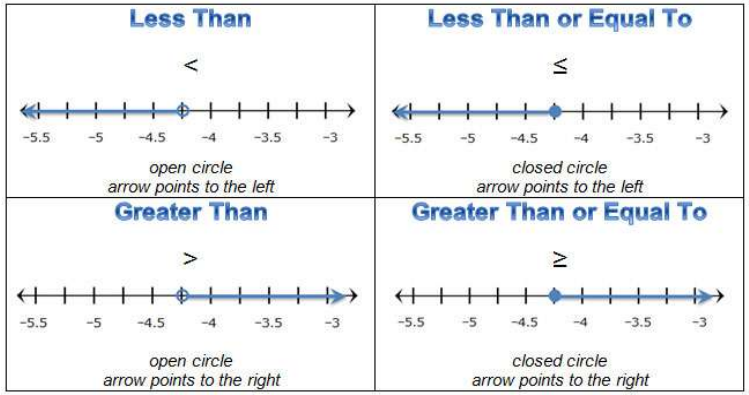
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**B9.2.2.1.4 Use the knowledge of simplifying and factorising expressions to solve real world problems.** **E.g.1** You purchased 10 items from a shopping plaza, and now you need plastic bags to carry them home. If each bag can hold only 3 items, how many plastic bags will you need to accommodate the10 items? **Solution**: We use simple algebraic formula $\frac{x}{y}$ to calculate the number of bags. x = Number of items purchased = 10 y = Capacity of 1 bag = 3 Hence, $\frac{10}{3}$ = 3.333 bags = 4 bags So, we need 4 shopping bags to carry 10 items.  	**Creativity and Innovation (CI)** **Digital Literacy (DL)**

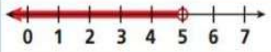
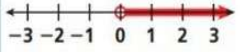
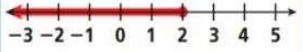
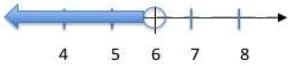
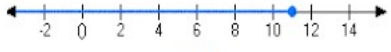
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**E.g.2** You have to buy two dozen of eggs priced at GH₵10, three loaves breads (each bread is GH₵5), and five bottles of juice (each bottle is GH₵8). How much money you will need to take to the grocery store? The prices are a = Price of two dozen eggs = GH₵10 b = Price of one bread = GH₵5 c = Price of one bottle of juice = GH₵8 => Money needed = a + 3b + 5c => Money needed = GH₵10 + 3(GH₵5) + 5(GH₵8) = GH₵10 + GH₵15 + GH₵40 = GH₵65 	**Communication and Collaboration (CC)**

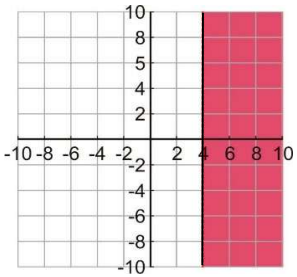
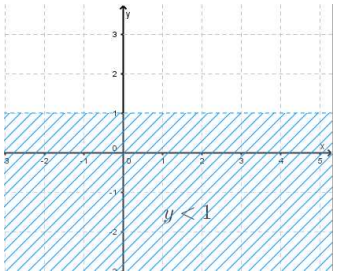
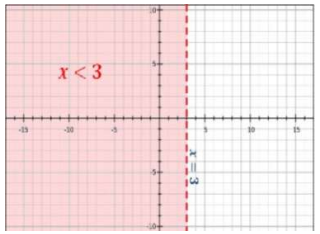
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.2.1Demonstrate an understanding of (i) change of subject (ii) substituting values to evaluate expressions, and (iii) factorize expressions that have simple binomial as a factor.	**E.g.3** The area of a rectangle is 72 cm^2. The length is twice its width. What is the length and width of the rectangle? Let "x" be the width and "2x" be the length. Length × Width = Area $x \times (2x) = 2x^2$ = Area $2x^2$ = Area $2x^2$ = 72 $\frac{2x^2}{2} = \frac{72}{2}$ $x^2 = 36$ x = 6 width = 6cm 2x = 2 x 6 = 12 So, the length is 12 cm	**Critical Thinking and Problem solving (CP)**

STRAND 2: ALGEBRA

Sub-strand 3 Variables and Equations

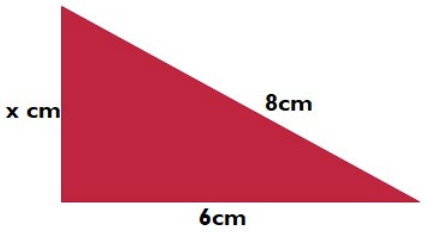
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<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr>
<td rowspan="3">B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including:
• solving inequalities
• verifying
• comparing
• graphing</td>
<td>B9.2.3.1.1 Solve single variable linear inequalities with rational coefficients.
<table>
<tr><td>i. $2x+7>\frac{5}{2}$</td><td>v. $\frac{1}{3}>x-\frac{4}{5}$</td></tr>
<tr><td>ii. $\frac{4}{5}-\frac{1}{5}x>\frac{2}{7}$</td><td>vi. $\frac{1}{2}(2x+3)\leq x+1$</td></tr>
<tr><td>iii. $\frac{3}{2}y-\frac{2}{5}<\frac{4}{5}$</td><td>vii. $x+\frac{1}{2}\geq-\frac{3}{2}$</td></tr>
<tr><td>iv. $\frac{1}{2}(5x-4)<x+\frac{11}{24}$</td><td>viii. $-\frac{2}{3}x+3\geq 0$</td></tr>
</table></td>
<td rowspan="3">Critical Thinking and Problem solving (CP)
Personal Development and Leadership (PL)
Digital Literacy (DL)</td>
</tr>
<tr><td>B9.2.3.1.2 Illustrate solution sets of linear inequalities on the number line</td></tr>
<tr><td>E.g.1 Illustrate and explain the inequality signs
 
</td></tr>
</table>

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr>
<td rowspan="3">B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including:
• solving inequalities
• verifying
• comparing
• graphing</td>
<td>E.g.2 Graph linear inequalities in one variable on a number line.
<table>
<tr><th>Word Phrase</th><th>Inequality</th><th>Solution Set</th></tr>
<tr><td>x is less than 5</td><td>$x<5$</td><td>  </td></tr>
<tr><td>a is greater than 0
a is more than 0</td><td>$a>0$</td><td>  </td></tr>
<tr><td>y is less than or equal to 2
y is at most 2</td><td>$y\leq 2$</td><td>  </td></tr>
<tr><td>m is greater than or equal to 3
m is at least 3</td><td>$m\geq 3$</td><td>  </td></tr>
</table>
E.g.3 Solve and graph linear inequalities on a number line.
i. $-3x-8>-26$
-3x – 8 > -26
+8 +8
-3x > -18
-3 -3
x < 6

ii. $2x-3\leq 19$
$2x\leq 22$
$\frac{2x}{2}\leq\frac{22}{2}$
$x\leq 11$

$x\leq 11$</td>
<td>Digital Literacy (DL)</td>
</tr>
</table>

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including: • **solving inequalities** • **verifying** • **comparing** • **graphing**	E.g.4 Solve and graph linear inequalities on a cartesian plane. i. $2x \geq 8$ $\frac{2x}{2} \geq \frac{8}{2}$ $x \geq 4$  ii. $5y + 3 < 6 + 2y$ $5y - 2y < 6 - 3$ $3y < 3$ $\therefore y < 1$  iii. $\frac{7x}{3} < 7 \rightarrow 3 \times \frac{7x}{3} < 7 \times 3$ $\frac{7x}{7} < \frac{21}{7}$ $x < 3$ 	**Digital Literacy (DL)**

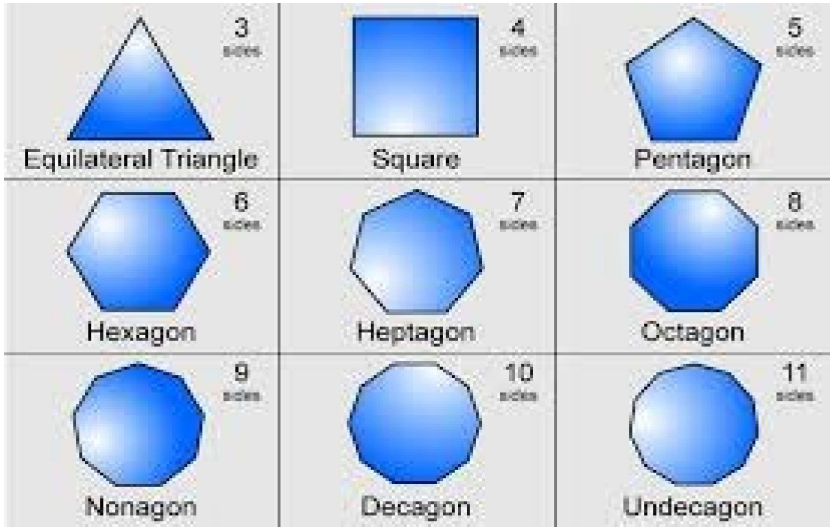
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including: • **solving inequalities** • **verifying** • **comparing** • **graphing**	**B9.2.3.1.3 Solve real-life problems involving linear equations and inequalities.**	**Critical Thinking and Problem solving (CP)**
	E.g.1 Solve real-life problems involving linear equations. i. A man has 260metres of fencing which he is going to put around a rectangular field which is 50metres wide. How long is the field? **Solution**: Since we need to find the length of the field, let x metresbe  the length. $x + 50 + x + 50 \; or \; 2(x + 50)$ But this expression is given as 260m $\therefore 2(x + 50) = 260$ $x + 50 = 130$ $x = 80\text{m}$	

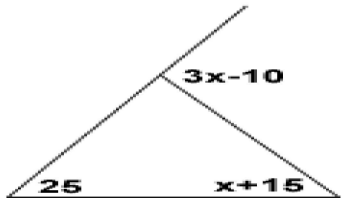
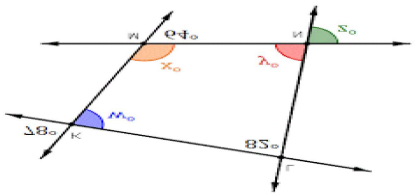
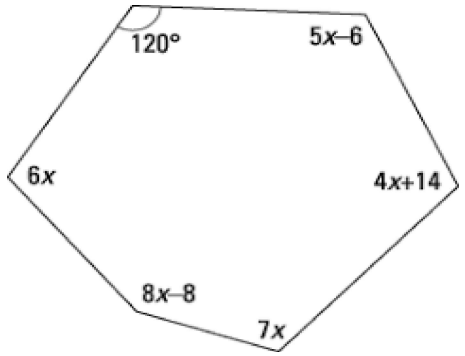
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including: • **solving inequalities** • **verifying** • **comparing** • **graphing**	ii. A man paid GH₵ 290 for 11 books. Some of the books were geography books, and the rest were history books. If each geography book cost GH₵ 30 and each history book cost GH₵20, how many geography books did he buy? Solution: i. Total cost of the books is GH₵290; total number of books is 11. ii. 1 geography book costs GH₵30; 1 history book costs GH₵20, Total cost of all the books is $30x + 20(11 - x) = 290$ $\therefore \mathbf{30x + 20(11 - x) = 290}$ $\mathbf{30x + 220 - 20x = 290}$ $\mathbf{10x + 220 = 290}$ $\mathbf{x = 7}$ Hence the number of geography books bought is 7.	**Critical Thinking and Problem solving (CP)**
	E.g.2 Solve real-life problems involving linear inequalities. i. Two sides of a triangle have lengths 6 cm and 8 cm. What is the length of the third side? **Note**: The sum of the lengths of the two sides of a triangle is greater than the length of the third side.	

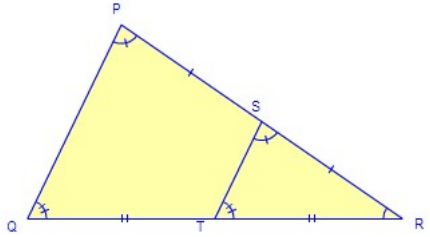
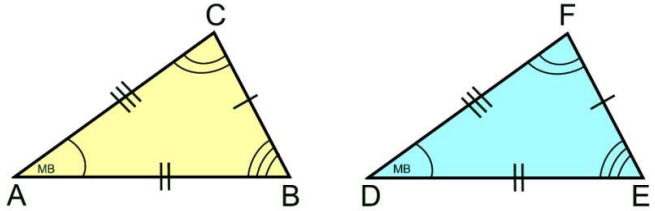
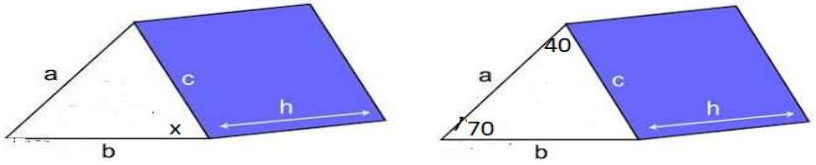
CONTENT STANDARDS	INDICATORS AND EXEMPLARS		CORE COMPETENCIES
B9.2.3.1 Demonstrate understanding of single variable linear inequalities with rational coefficients including: • **solving inequalities** • **verifying** • **comparing** • **graphing**	If the third side is $x$cm long then, $6 + 8 > x$ giving $x < 14$. Also, $6 + x < 8\ giving\ x > 2$. [Also, $8 + x > 6$ whichgives$x > -2$. Hence, $2 < x < 14$. that is, the third side has length between 2cm and 14cm. ii. A student scores 70 and 76 marks in two tests. How many marks must she score in the third test to be put in Grade A if all students scoring an average of 80 or higher in three tests are put in grade A?		**Critical Thinking and Problem solving (CP)**

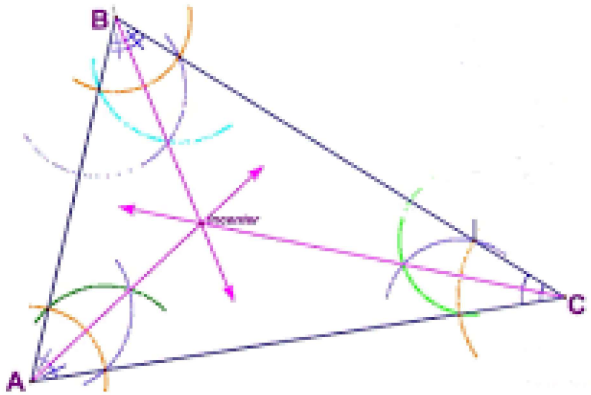
STRAND 3: GEOMETRY AND MEASUREMENT

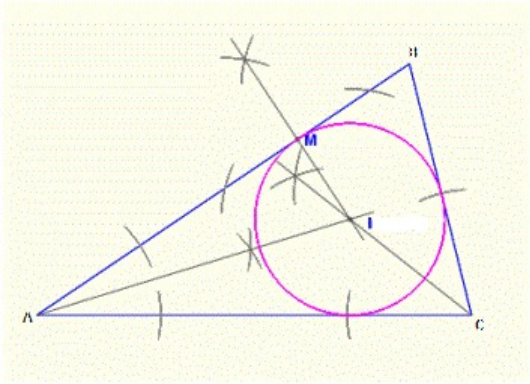
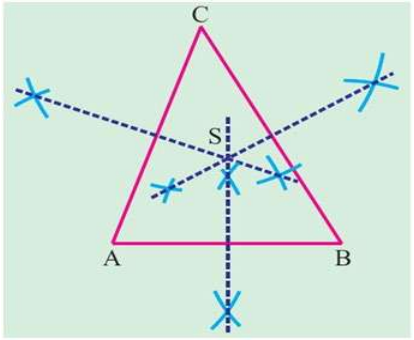
Sub-strand 1 Shapes and Space

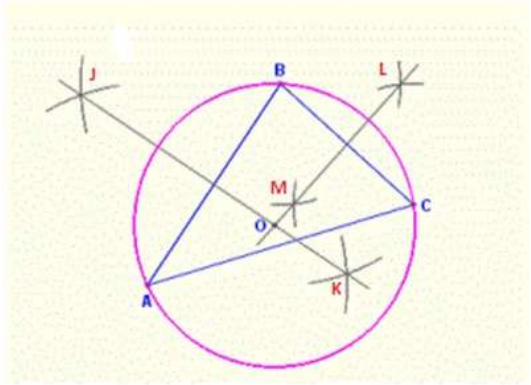
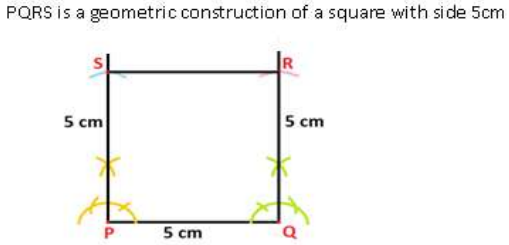
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.1 Apply the properties of angles at a point, angles on a straight line, vertically opposite angles, corresponding, angles to` solve problems.	**B9.3.1.1.1Derive the formula for calculating the sum of angles in any polygon and use this to calculate the value of missing angles in polygons** **E.g.1.** Identify and name the various polygons such as a triangle, quadrilaterals, pentagons, and hexagons, etc. 	**Communication and Collaboration (CC)**

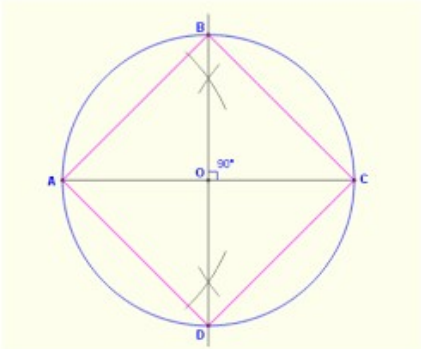
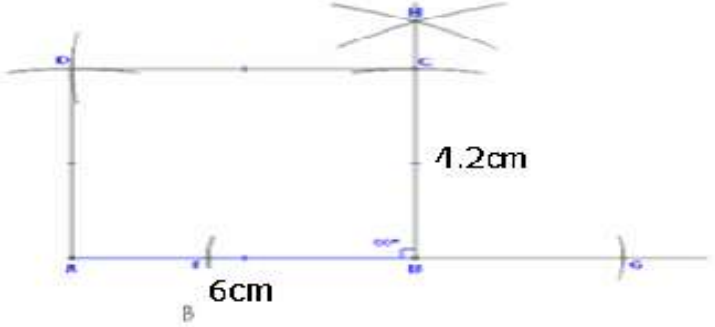
CONTENT STANDARDS	INDICATORS AND EXEMPLARS		CORE COMPETENCIES
B9.3.1.1 Apply the properties of angles at a point, angles on a straight line, vertically opposite angles, corresponding, angles to` solve problems.	E.g.2. Derive and use the formula (n - 2) × 180 ° and calculate the value of x (interior and angles of a triangle).		**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)**
	E.g.3. Derive and use the formula (n - 2) × 180 ° and calculate the interior angles of a quadrilateral.		
	E.g.4. Derive and use the formula (n - 2) × 180 ° and calculate the interior angles of polygons, pentagons, hexagons, etc. (i) Find the value of x and the various angles in the hexagon.		

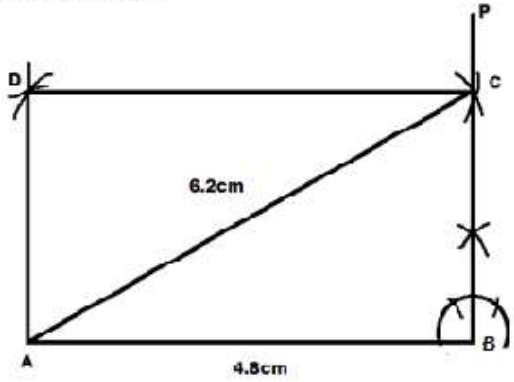
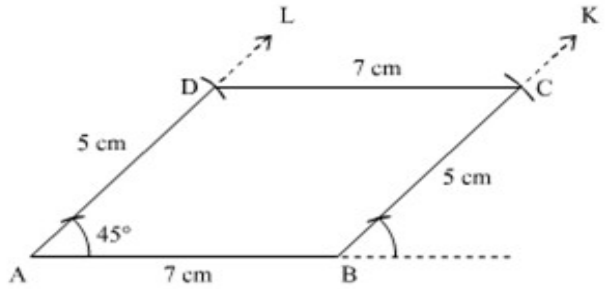
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.1 Apply the properties of angles at a point, angles on a straight line, vertically opposite angles, corresponding, angles to` solve problems.	**B9.3.1.1.2 Identify similar and congruent triangles and use the knowledge to solve related problems** E.g.1. Recognise similar triangles and solve for the values of the indicated angles in the diagram below: 	**Critical Thinking and Problem solving (CP)** **Creativity and Innovation (CI)**
	E.g.2. Recognise congruent triangles and solve for the values of the indicated angles in the diagram below:  $\Delta ABC \cong \Delta DEF$	
	E.g. 3. Determine the value of x (using knowledge in similarity and congruency). 	

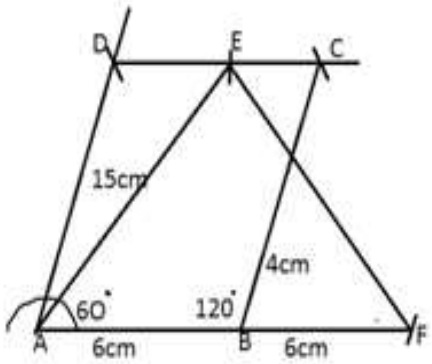
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	**B9.3.1.2.1 Draw inscribed and circumscribed circles for triangles under given conditions** E.g.1: Use a pair of compasses and a ruler to construct a triangle (say ΔABC) under a given condition and locate the incentre of the triangle (the incentre is the point of concurrency of the three angle bisectors of a triangle); measure the shortest distance from the incentre to the line segments AB, AC and BC. What do you observe about the lengths? 	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	E.g..2: Use a pair of compasses and a ruler to construct a triangle (say ABC) under a given condition, bisect at least any two angles (ΔBAC and ΔBCA); locate the intersection of the two angle bisectors (L) and draw a locus of points equidistant from the fixed point (L) to touch the edges of the triangle. 	**Creativity and Innovation (CI)** **Digital Literacy (DL)**
	E.g.3: Construct a triangle (say ABC); bisect all three sides (i.e. line segments AB, AC and BC); locate the intersection (circumcentre) of the three perpendicular bisectors (S); Measure the distance from the intersecting centre (S) to points A, B and C. What do you observe about the lengths? 	**Personal Development and Leadership (PL)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	E.g.4: Perform a geometric construction of a triangle (say ABC) under a given condition; bisect at least any two sides (AB and BC); locate the intersection of the two perpendicular bisectors (O) and draw a locus of points equidistant from the fixed point (O) to circumscribe the triangle.  	**Creativity and Innovation (CI)**
	B9.3.1.2.2 Construct parallelograms (i.e. square, rectangle, rhombus) under given conditions. E.g.1: Perform geometric construction of a square with a given side.  	**Communication and Collaboration (CC)**

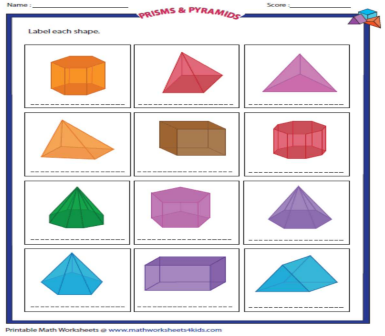
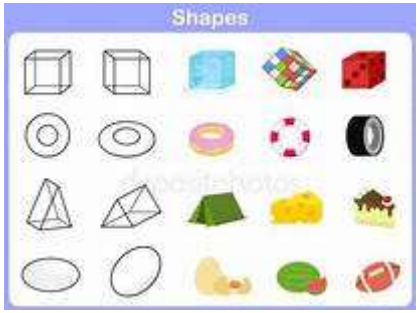
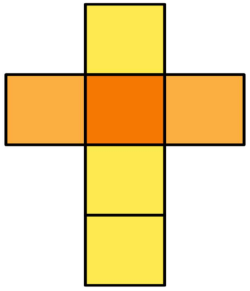
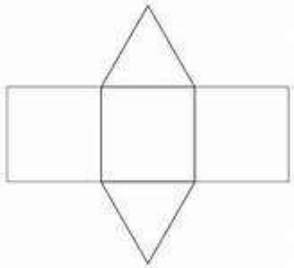
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	E.g.2: Construct a square ABCD with $\overrightarrow{AB}$=6.5cm. Measure and record the diagonal of the square. E.g.3: Perform geometric construction of a square with a given diagonal. Construct the square ABCD with AC = 10cm. What is the length of the sides? 	**Critical Thinking and Problem solving (CP)**
	E.g.4: Perform geometric construction of a rectangle with given side. Construct rectangle ABCD such that AB=6cm and BC=4.2cm 	

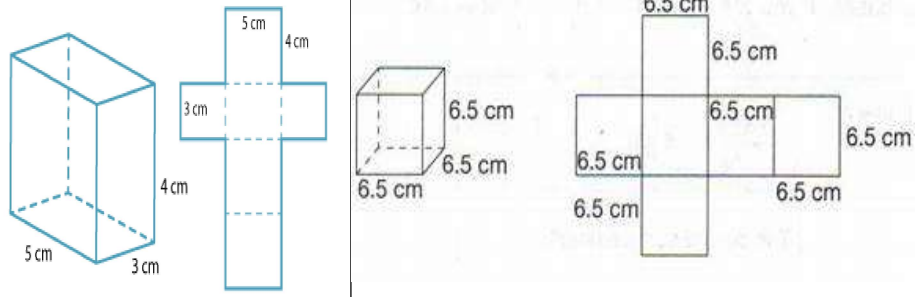
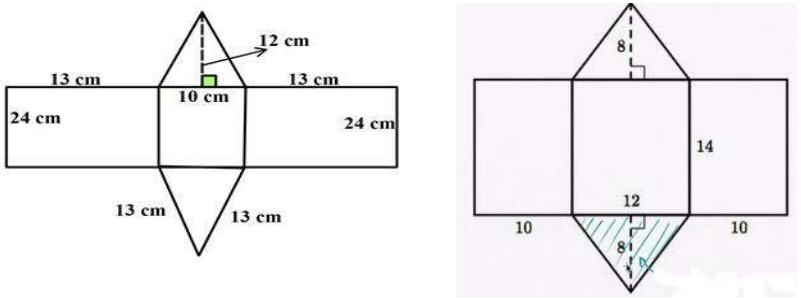
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	E.g.5: Perform geometric construction of a rectangle with a given side and diagonal. Construct a rectangle ABCD with length AB = 4.8cm and diagonal AC= 6.2cm  	**Critical Thinking and Problem solving (CP)**
	E.g.6: Perform geometric construction of a parallelogram with given sides and given angle(s). Construct the parallelogram ABCD such that the line segments AB=7cm and AD=5cm, and < DAB=45°  	**Creativity and Innovation (CI)**

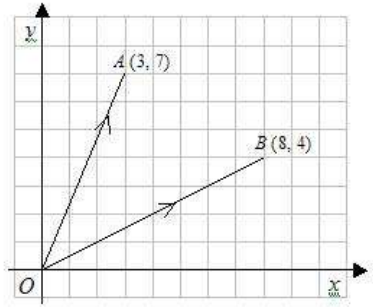
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.1.2 Construct inscribed and circumscribed triangles and parallelograms with given dimensions.	E.g.7: Perform geometric construction of regular compound plane shapes with given sides and angles. ABCD is a parallelogram such that <DAB=60°, <CBA = 120°, AB=6cm. AEF is an isosceles triangle, such that AE=FE= 15cm, B is the midpoint of AF=12cm.  	**Digital Literacy (DL)**

STRAND 3: GEOMETRY AND MEASUREMENT

Sub-strand 2 Measurement

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B.9.3.2.1 Derive the formulas for determining the surface area of prisms (i.e. cuboid and triangular prism) and use to solve problems.	**B9.3.2.1.1 Identify cuboids and triangular prisms; draw their nets to construct the 3-D shapes and use it to determine the surface area.** E.g.1 Sort out shapes that are triangular prisms and cuboids.   E.g.2 Identify each of the nets below:  A net of……..  A net of……..	**Critical Thinking and Problem solving (CP)** **Creativity and Innovation (CI)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B.9.3.2.1 Derive the formulas for determining the surface area of prisms (i.e. cuboid and triangular prism) and use to solve problems.	E.g.3 Measure and find the area of each of the sections in the net and adding all together to give the surface area.	**Critical Thinking and Problem solving (CP)**
	B9.3.2.1.2 Use the net of a cuboid to determine its surface area. E.g.1 Find the surface area of each of the cuboids	
		
	B9.3.2.1.3 Use the net of a triangular prism to determine its surface area. E.g.1Find the surface area of each of the triangular prims.	
		

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B.9.3.2.1 Derive the formulas for determining the surface area of prisms (i.e. cuboid and triangular prism) and use to solve problems.	**B9.3.2.1.4 Express points in the Cartesian plane as position vectors**	**Creativity and Innovation (CI)**
	E.g.1 Identity the following using the diagram below: (i) the origin (ii) the position vector If $a = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$, then the coordinates of A will be (3, 7). Similarly, if $b = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$, then coordinates of B will be (8, 4) E.g.2 Draw and write the position vectors of the following with 0 as the origin: (i) M(2,3) (ii) N(-1,2) 	

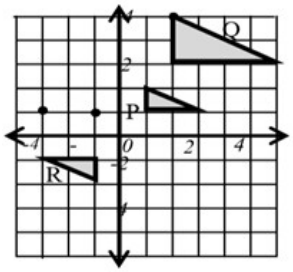
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.2.2 Solve problems involving bearings and addition/subtraction of vectors	**B9.3.2.2.1 Show an understanding of parallel vectors and perpendicular vectors** E.g.1 Investigate conditions for parallel vectors and perpendicular vectors. E.g.2 Use the result from the investigation to solve the following questions: (i) Find the value(s) of x, if the vectors $\begin{pmatrix}3x\\2\end{pmatrix}$ and $\begin{pmatrix}6\\x\end{pmatrix}$ are parallel. (ii) Which of the vectors is perpendicular to $\begin{pmatrix}3\\4\end{pmatrix}$ (a) $\begin{pmatrix}-3\\4\end{pmatrix}$ (b) $\begin{pmatrix}-3\\-4\end{pmatrix}$ (c) $\begin{pmatrix}-4\\3\end{pmatrix}$ (d) $\begin{pmatrix}-4\\-3\end{pmatrix}$	**Digital Literacy (DL)**
	B9.3.2.2.2 Apply the triangular and parallelogram laws of addition to resolve vectors E.g.1 Deduce the triangle law of vector addition. $\overrightarrow{AB}$ +$\overrightarrow{BC}$ = $\overrightarrow{AC}$ Where ABC are points in the 0xy plane. E.g.2 The vertices of a triangle are P(1,-3), Q(7,5) and R(-3,5) (i) Express $\overrightarrow{PQ}, \overrightarrow{QR}$, and $\overrightarrow{PR}$ as column vectors. (ii) Show that triangle PQR is an isosceles. (iii) Find the equation of the line$\overrightarrow{PR}$. Eg3 Investigate the parallelogram law of vector addition. Eg.4P,Q,R,S is a parallelogram whose vertices are P (x ,y), Q (5,7), R(2,4) and S(1,3) (i) Find $\overrightarrow{PQ}$, and$\overrightarrow{SR}$ hence find the values of x and y.	**Creativity and Innovation (CI)**

STRAND 3:GEOMETRY AND MEASUREMENT

Sub-strand 3 Position and Transformation

<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr><td rowspan="2">B9.3.3.1 Demonstrate understanding of how to perform an enlargement on a geometrical shape given a scale factor and describe the properties of the image under the transformation (i.e. congruence, similarity, etc.)</td><td colspan="2">B9.3.3.1.1 Know examples of situations in everyday life that depict enlargement situations in everyday life.
E.g. 1. Know examples of situations that relate to enlargement situations in everyday life and the nature of movements – vertical and horizontal.</td><td rowspan="2">Creativity and Innovation (CI)

Digital Literacy (DL)</td></tr>
<tr><td>  </td><td> 
his parents had taught at the Defence Services Academy, Myanmar's equivalent of America's West Point. Egress thus acted as a bridge. It engineered the first official meeting between Thein Sein and Miss Suu Kyi in 2011. Tunisia's singular success is partly due to the efforts of its national dialogue quartet, a disparate collection of unionists, employers, lawyers and human-rights activists. They recently won the Nobel peace prize.
Good neighbours and sponsors</td></tr>
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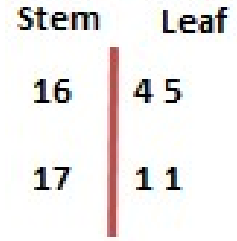
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.3.3.1 Demonstrate understanding of how to perform an enlargement on a geometrical shape given a scale factor and describe the properties of the image under the transformation (i.e. congruence, similarity, etc.)	**B9.3.3.1.2 Understand enlargement and identify real-life situations involving enlargement.** E.g. 1. Draw an enlargement of shapes using a given scale factor. i. State the single transformation that maps triangle P onto Q. ii. State the single transformation that maps triangle P onto R.  iii. Investigate the characteristics of enlargements under the following conditions of scale factor: • if the scale factor (K) is negative • if the scale factor (K) is greater than 1 or less than –1 • if the scale factor (K) is between –1 and 1 (i.e., fraction) E.g. 2. Using an object, and its image, determine the scale factor in a transformation?	**Creativity and Innovation (CI)**
	B9.3.3.1.3 Investigate the concept of congruent and similar shapes E.g. 1. Using multiple and varied examples of enlargement on a coordinate plane, verify congruent and similar shapes using their properties.	

STRAND 4: HANDLING DATA

Sub-strand 1: Data

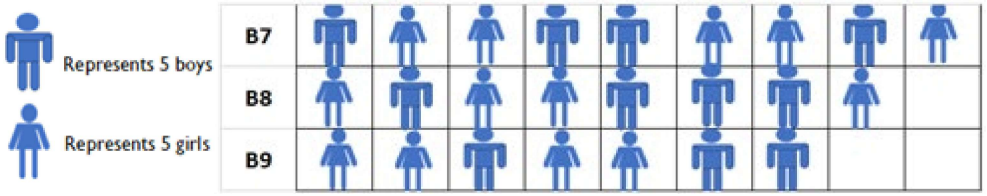
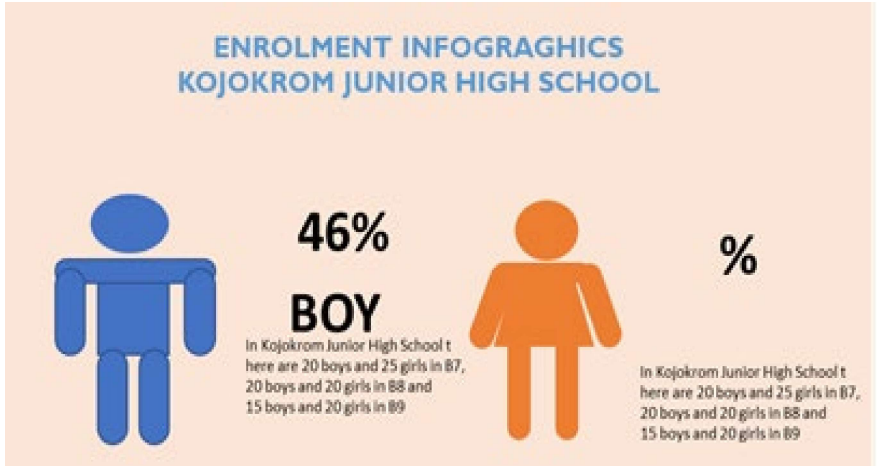
CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.1Select, justify, and use appropriate methods of collecting data (grouped/ungrouped), use the data to construct and interpret frequency tables and histogram and use it to determine the mode and to solve and/or pose problems.	**B9.4.1.1.1Select and justify a method to collect data (quantitative and qualitative) to answer a given question.** E. g. 1 Discuss and decide (i) from where/whom to collect data for the studies presented below; (ii) which data collection methods to use; and (iii) justify the choices for (i) and (ii) above Areas of study are described as follows: a. Musa has started a book club for Ayisha and her friends. He wants Ayisha to find out books that are most popular among her friends. **b.** Find the most common mode of travel by learners in Oyoko Junior and Senior High Schools.	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)**
	B9.4.1.1.2.Organise data (grouped/ungrouped) present it in frequency tables, line graphs, pie graphs, bar graphs and/or pictographs (representations include infographics, waffle diagrams, box and whisker plots and stem and leaf plots) and analyse it to solve and/or pose problems. E.g. 1 -Thirty bulbs were life-tested and their lifespan to the nearest hour are as follows: 167 171 179 167 171 165 175 179 169 171 177 169 171 177 173 165 175 167 174 177 172 164 175 179 179 174 174 168 171 168	

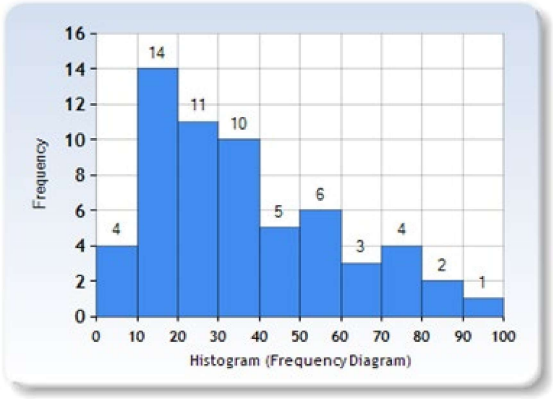
<table>
<tr><th>CONTENT STANDARDS</th><th>INDICATORS AND EXEMPLARS</th><th>CORE COMPETENCIES</th></tr>
<tr>
<td>B9.4.1.1Select, justify, and use appropriate methods of collecting data (grouped/ungrouped), use the data to construct and interpret frequency tables and histogram and use it to determine the mode and to solve and/or pose problems.</td>
<td>
i. Present the raw data in a frequency table by completing the table below:
<table>
<tr><th>Lifespan of Bulbs (hours)</th><th>Tally</th><th>Frequency</th></tr>
<tr><td>164 - 167</td><td></td><td></td></tr>
<tr><td>168 – 171</td><td></td><td></td></tr>
<tr><td>172 - 175</td><td></td><td></td></tr>
<tr><td>176 – 179</td><td></td><td></td></tr>
</table>
ii. What is the modal group? Justify your decision for that choice.

iii. Complete the stem and leaf plots below to display the raw data.
Use the plot to solve the following problems.
a. Find the range of the lifespan of bulbs
b. What is the mode lifespan?
c. What is the median lifespan?
d. What other problems can you pose?


</td>
<td>Critical Thinking and Problem solving (CP)

Creativity and Innovation (CI)</td>
</tr>
</table>

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.1Select, justify, and use appropriate methods of collecting data (grouped/ungrouped), use the data to construct and interpret frequency tables and histogram and use it to determine the mode and to solve and/or pose problems.	E.g. 2 The pictograph below describes the number of boys and girls in each class in Kojokrom Junior High School 	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)**
		
	i. What is the percentage of boys and of girls in the school? ii. Use your answers in (i) to represent the data by copying and completing the following infographic	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.1Select, justify, and use appropriate methods of collecting data (grouped/ungrouped), use the data to construct and interpret frequency tables and histogram and use it to determine the mode and to solve and/or pose problems.	**B9.4.1.1.3 Use a histogram to determine the mode of a given data to solve and/or pose real life cases.** E.g. 1- The waiting times, x minutes, for 60 patients at a certain clinic are as follows 25 12 53 8 26 5 19 73 67 18 87 42 6 21 14 19 12 15 13 36 36 16 72 36 13 37 11 51 39 32 30 47 6 22 68 25 98 23 45 22 7 9 26 35 27 48 58 56 29 20 32 62 80 41 58 17 54 15 14 74 i. Construct a frequency table using class intervals 0 – 10.5; 10.5 – 20.5; 20.5– 30.5, and so on. ii. Construct a frequency table using class intervals $0 < x \leq 10; 10 < x \leq 20;$ $20 < x \leq 30$, and so on. iv. Draw a histogram and find the modal class. 	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.2 Select, justify, and use appropriate methods of collecting data (quantitative and qualitative), organise and analyse the data (grouped/ungrouped) to interpret the results using the descriptive statistics (measures of central tendency and range).	**B9.4.1.2.1 - Select a method for collecting data (quantitative and qualitative), taking into consideration how bias (use of language, ethics, cost, time and timing, privacy or cultural sensitivity) may influence data.** E. g. 1 Suppose in a school survey form the following question was asked: **Overall, don't you think the teaching of mathematics is very good?** The designer of the survey form has a **bias** for the methodology used in math lessons and the bias influences how the question was written. The **language** used in writing the question may lead people to just answer **yes** or **no**. A better question would be: Overall, how will you rate the teaching of mathematics? ◯ **Very poor** ◯ **Poor** ◯ **Fair** ◯ **Good** ◯ **Very Good**	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**
	E.g. 2 Ama Mereku in B9 wants to write an article for their school magazine on **sport-related injuries.** The responses for the survey question stated below were **collected** from **only the schools' football team**. The influencing factors in this survey question are: **time** and **bias**. Football is a contact sport. The chances are that the answers from her targeted respondents will be high in favour of injuries and thus negatively affect the conclusion/report. In order to report **accurately** on sport-related injuries Ama needs to ask **more** people (**time needed)** who participate in a **variety of sports,** including **contact and non-contact** sports **(e.g. athletics tennis, volleyball,** and so on).	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.2 Select, justify, and use appropriate methods of collecting data (quantitative and qualitative), organise and analyse the data (grouped/ungrouped) to interpret the results using the descriptive statistics (measures of central tendency and range).	E.g. 3Learners in B9 are asked by their physical education teacher to complete a survey related to "**Overall Physical Health**". One question on the survey form is; ***What is your current body weight?*** Identify the influencing factor in the survey and provide a solution.	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**
	E.g. 4 - Suppose you tell your classmates that the response to the question in the Class Survey Question Form is to help you plan remedial classes. ***What is your worst subject?*** If you then use the information collected **to write an article for the school magazine** how would your actions be described and how would that influence future surveys you conduct?	
	E.g. 5 -Suppose in a survey questionnaire you wanted to know the favourite method of cooking pork and you asked: Please tick the box against your favourite method of cooking pork ☐ Boiling ☐ Grilling ☐ Frying Please tick the box against your favourite method of cooking pork **(Optional)** ☐ Boiling ☐ Grilling ☐ Frying This question does not apply to everyone because some people do not eat pork (i.e. the question is not culturally sensitive.) A better question would be **If** you eat pork please name the favourite method you cook it. ☐ Boiling ☐ Grilling ☐ Frying	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.2 Select, justify, and use appropriate methods of collecting data (quantitative and qualitative), organise and analyse the data (grouped/ungrouped) to interpret the results using the descriptive statistics (measures of central tendency and range).	OR **If** you eat pork please name the favourite method you cook it. ☐ Boiling ☐ Grilling ☐ Frying **B9.4.1.2.2 Organise and analyse data and interpret the results using the descriptive statistics (i.e. minimum, maximum, measures of central tendency and range) to answer a given question.** Refer to E.g. 1 of **B9.4.1.1.2** and find (minimum, maximum, measures of central tendency and range) i. The minimum lifespan, to the nearest hour, of the bulbs tested. ii. The maximum lifespan, to the nearest hour, of the bulbs tested. iii. The range of the data collected from the life-testing. iv. What is the mean lifespan of the bulbs? v. What is the median of the lifespan of the bulbs? vi. What is the mode of the lifespan of the bulbs? vii. When placing an order for the bulbs tested to sell in your shop, which of them will you consider buying?	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.1.2 Select, justify, and use appropriate methods of collecting data (quantitative and qualitative), organise and analyse the data (grouped/ungrouped) to interpret the results using the descriptive statistics (measures of central tendency and range).	**B9.4.1.2.3Demonstrate the effect on the mean, median, and mode when extreme data is included in a data set** E.g.1 Refer to E.g. 1 of B9.4.1.1.2. i. Find the mean of the data, if one of the bulbs is replaced with a new bulb with lifespan of 300 hours, find the new mean of the bulbs and compare it to the original mean ii. In small groups, find the mean of the data, if the lifespan of one of the bulbs tested was 70 hours, and compare it to the original mean. iii. Continue to replace the values of the lifespan in the data *with extreme values* (small and large), calculate the mean, median, and mode and discuss the findings.	**Critical Thinking and Problem solving (CP)** **Communication and Collaboration (CC)** **Creativity and Innovation (CI)** **Personal Development and Leadership (PL)** **Digital Literacy (DL)**

STRAND 4: HANDLING DATA

Sub-strand 2: Chance or Probability

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
B9.4.2.1Identify the sample space for a probability experiment involving two dependent events and express the probabilities of given events as fractions, decimals, percentages and/or ratios to solve problems.	**B9.4.2.1.1.Perform a probability experiment involving two dependent events e.g. drawing coloured bottle tops from a bag without replacement** E.g. 1In an experiment, Anita was asked to pick one bottle top, in three trials, from a bag which contains 3 red, 2 green and 1 pink bottle tops **without replacement**. i. List the elements of the sample space of the events. ii. Does the occurrence of the one trial affect the occurrence of the other trials?	**Critical Thinking and Problem solving (CP)** **Personal Development and Leadership (PL)**
	B9.4.2.1.2. Express the probabilities of the events as fractions, decimals, percentages and/or ratios; e.g. using a tree diagram, table or another graphic organiser	
	E.g. 1 Draw a probability tree diagram for the experiment in B9.4.2.1.1, E.g. 1. Express the probabilities of the events (on their respective branches) as decimals, percentages and ratios.	
	E.g. 2: i. Consider the experiment of drawing two Aces (in two trials) in a standard deck of cards without replacement. ii. Calculate the probability of **each** trial and express the probabilities of the events as decimals, percentages and ratios.	

CONTENT STANDARDS	INDICATORS AND EXEMPLARS	CORE COMPETENCIES
	E.g. 3: i. Consider the experiment of drawing an Ace and a Jack (in two trials) in a standard deck of cards without replacement. ii. Calculate the probability of **each** trial and express the probabilities of the events as decimals, percentages and ratios.	

APPENDIX: UNPACKING THE CORE COMPETENCES OF THE STANDARDS-BASE CURRICULUM

NATIONAL COUNCIL FOR CURRICULUM AND ASSESSMENT (NaCCA)

CORE COMPETENCIES AND SUBSKILLS OF THE COMMON CORE PROGRAM (CCP)

1. COMMUNICATION AND COLLABORATION (CC)

B7-B10		
CC7: LISTENING	**CC8: PRESENTING**	**CC9: TEAMWORK**
CC7.1: Identify words or sentences in context appropriately	**CC8.1**: Speak clearly and explain ideas. Share a narrative or extended answer while speaking to a group	**CC9.1**: Demonstrate behaviour and skills of working towards group goals
CC7.2: Interpret correctly and respond to non- verbal communication such as facial expressions, cues and gestures	**CC8.2**: Explain ideas in a clear order with relevant detail, using correct construction and structure of speech	**CC9.2**: Understand and use interpersonal skills
CC7.3: Provide feedback in areas of ideas, organisation, voice, word choice and sentence fluency in communication	**CC8.3**: Apply appropriate diction, and structure sentences correctly for narrative, persuasive, imaginative and expository purposes	**CC9.3**: Understand roles during group activities
CC7.4: Identify underlying themes, implications and issues when listening	**CC8.4**: Anticipate different responses from the audience and plan for them	**CC9.4**: Help group work on relevant activities
CC7.5: Identify and analyse different points of views of speaker	**CC8.5**: Vary the level of detail and the language used when presenting to make it appropriate to the audience	**CC9.5**: Appreciate the importance of including all team members in discussions and actively encourage contributions from them
		CC9.6: Ability to work with all group members to complete a task successfully
		CC9.7: Effectively perform multiple roles within the group
		CC9.8: Demonstrate an awareness of the wider team dynamics and work to minimise conflicts in the team

2. CRITICAL THINKING AND PROBLEM SOLVING (CP)

B7-B10	
CP5: CRITICAL THINKING	**CP6: PROBLEM SOLVING**
CP 5.1: Ability to combine Information and ideas from several sources to reach a conclusion	**CP 6.1:** Ability to effectively define goals towards solving a problem
CP 5.2: Analyse and make distinct judgement about viewpoints expressed in an argument	**CP 6.2:** Ability to explain plans for attaining goals
CP 5.3: Create simple logic trees to think through problems	**CP 6.3:** Identify important and appropriate alternatives
CP 5.4: Generate hypothesis to help answer complex problems	**CP 6.4:** Ability to identify important and appropriate criteria and use them to evaluate available alternatives
CP 5.5: Effectively evaluate the success of solutions used in an attempt to solve a complex problem	**CP 6.5:** Ability to select alternative(s) that adequately meet selected criteria
CP 5.6: Demonstrate a thorough understanding of a generalised concept and facts specific to task or situation	**CP 6.6:** Preparedness to recognise and explain results after implementation of plans
CP 5.7: Provide new insight into controversial situation or task	**CP 6.7:** Implement strategies with accuracy
CP 5.8: Identify and prove misconceptions about a generalised concept or fact specific to a task or situation	
CP 5.9: Identify and explain a confusion, uncertainty, or a contradiction surrounding an event	
CP 5.10: Develop and defend a logical plausible resolution to a confusion, uncertainty or contradiction surrounding an event	

3. PERSONAL DEVELOPMENT AND LEADERSHIP (PL)

B7-B10	
PL5: PERSONAL DEVELOPMENT	**PL6: LEADERSHIP**
PL5.1: Understandingoneself (strengths, weaknesses, goals and aspirations),in reacting and adjusting to novel situations	**PL6.1:** Ability to serve group members effectively
PL5.2: Demonstrate a sense of belongingness to a group	**PL6.2:** Division of tasks into solvable units and assigning group members to task units
PL5.3: Recognise one's emotional state and their preparedness to apply emotional intelligence	**PL6.3:** Ability to manage time effectively

PL5.4: Ability to understand one's personality traits	**PL6.4:** Ability to manage and resolve conflicts
PL5.5: Desire to accept one's true self and overcome weaknesses	**PL6.5:** Ability to monitor team members to ascertain progress
PL5.6: Ability to set and maintain personal standards and values	**PL6.6:** Ability to mentor peers
	PL6.7: Actively promote effective group interaction and the expression of ideas and opinions in a way that is sensitive to the feelings and background of others
	PL6.8: Actively assist group identify changes or modifications necessary in the group activities and work towards carrying out those changes

4. CULTURAL IDENTITY AND GLOBAL CITIZENSHIP (CG)

B7-B10	
CG5: CULTURAL IDENTITY	**CG6: GLOBAL CITIZENSHIP**
CG5.1: Show a strong sense of belongingness to one's culture	**CG6.1:** Understanding of influences of globalisation on traditions, languages and cultures
CG5.2: Develop and exhibit ability to defend one's cultural beliefs, practices and norms	**CG6.2:** Recognise resistance to global practices that are inimical to our culture
CG5.3: Develop and express respect, recognition and appreciation of others' cultures	**CG6.3:** Know the global discourse about the roles of males and females
CG5.4: Develop and exhibit a sense of cultural identity	**CG6.4:** Exhibit a sense of nationality and global identity
CG5.5: Adjust to the demands of customs, traditions, values and attitudes of society	

5. CREATIVITY AND INNOVATION (CI)

B7-B10	
CI5: KNOWLEDGE, UNDERSTANDING, SKILLS AND STRATEGIES	**CI6: REFLECTION AND EVALUATION**
CI 5.1: Examine alternatives in creating new things	**CI 6.1:** Exhibit strong memory, intuitive thinking, and respond appropriately
CI 5.2: Ability to merge simple/complex ideas to create novel situations or things	**CI 6.2:** Ability to reflect on approaches to creative tasks and evaluate the effectiveness of tools used
CI 5.3: Identification of requirements of a given situation and justification of more than one creative tool that will be suitable	**CI 6.3:** Ability to select the most effective creative tools for work, and give reasons for the choice
CI 5.4: Ability to visualise alternatives, see possibilities, and identify problems and challenges	**CI 6.4:** Imagining and seeing things in a different way
CI 5.5: Ability to try new alternatives and different approaches	**CI 6.5:** Anticipate and overcome difficulties relating to taking initiatives
CI 5.6: Understand and use analogies and metaphors	**CI 6.6:** Being open-minded, adapting and modifying ideas to achieve creative results
CI 5.7: Putting forward constructive comments, ideas, explanations and new ways of doing things	**CI 6.7:** Look and think about things differently and from different perspectives
	CI 6.8: Recognise and generalise information and experience; search for trends and patterns
	CI 6.9: Interpret and apply learning in new contexts
	CI 6.10: Reflect on work and explore the thinking behind thoughts and processes

6. DIGITAL LITERACY (DL)

B7-B10	
DL5: PHOTO-VISUAL AND INFORMATION LITERACY	**DL6: SOCIO-EMOTIONAL AND REPRODUCTION LITERACY**
DL5.1: Ability to ascertain when information is needed and be able to identify, locate, evaluate and effectively use it to solve a problem	**DL 6.1:** Understand the sociological and emotional aspects of cyberspace
DL5.2: Ability to recognise and avoid traps in cyberspace	**DL 6.2:** Create a meaningful and original piece of work, or its interpretation by integrating existing information
DL5.3: Ability to find and utilise digital content	**DL6.3:** Use digital tools to create novel things
DL5.4: Ability to construct knowledge from a non-linear hyper-textual navigation	**DL6.4:** Adhere to behavioural protocols that prevail in cyberspace
DL5.5: Evaluate the quality and validity of information	**DL6.5:** Recognition of societal issues emanating from the use of digital technologies
DL5.6: Preparedness to make better decisions using available information	**DL6.6:** Knowledge and recognition of ethical use of information

Please note these inclusivity issues

The core competencies outlined in this document must be assessed taking into consideration people with special needs (physical disabilities, learning disabilities, etc.). Consider the use of realia for visual and visually challenged learners.

A system of creating alternatives for tasks must also be adopted.

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